43.4: Disruptions in immune system function can elicit or exacerbate disease

Although adaptive immunity protects against many pathogens, it is not fail-safe. Here we'll first examine the disorders and diseases that arise when adaptive immunity is blocked or misregulated. We'll then turn to some of the evolutionary adaptations of pathogens that diminish the effectiveness of adaptive immune responses in the host.

Exaggerated, Self-Directed, and Diminished Immune Responses

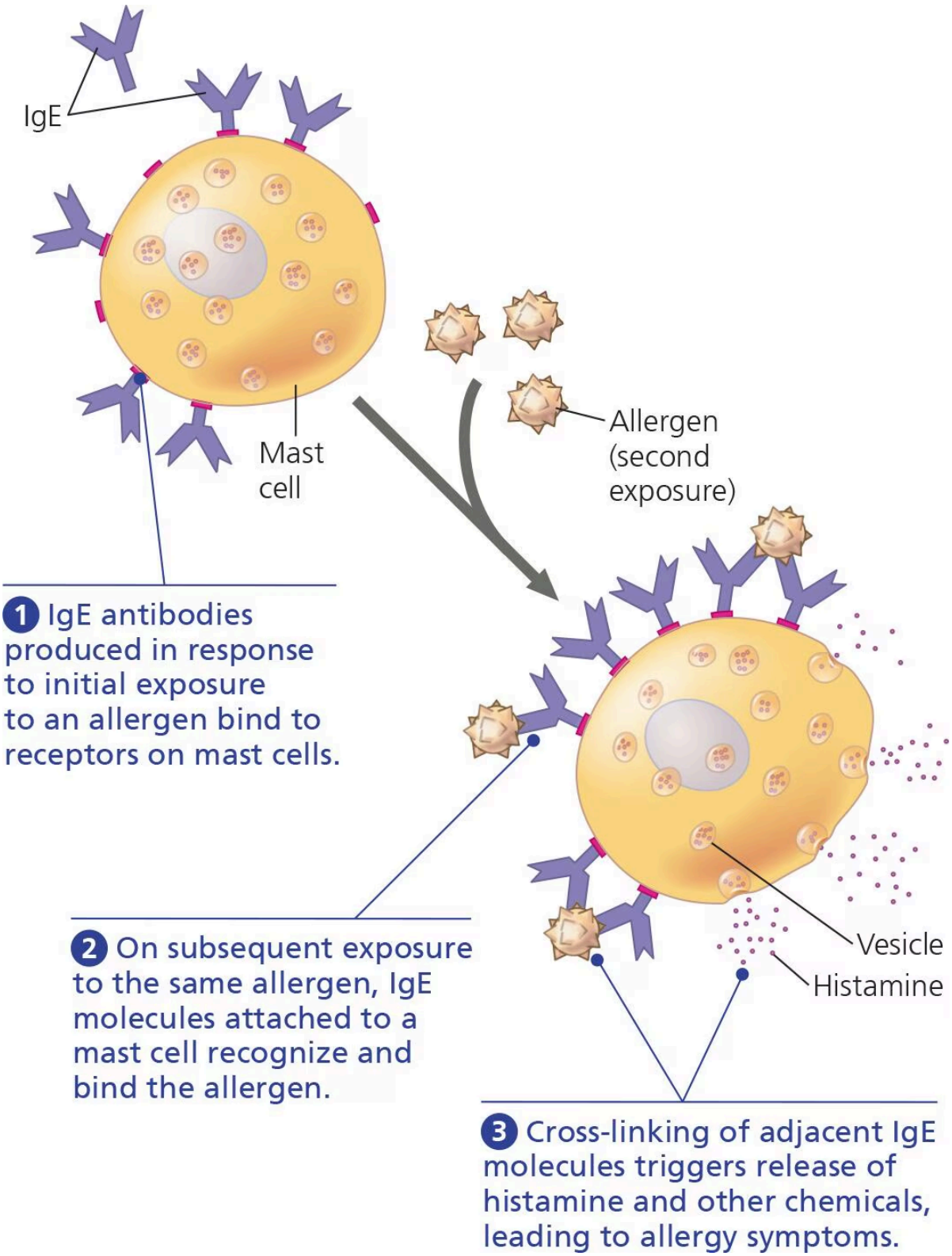
The highly regulated interplay among lymphocytes, other body cells, and foreign substances generates an immune response that provides extraordinary protection against many pathogens. When allergic, autoimmune, or immunodeficiency disorders disrupt this delicate balance, the effects are frequently severe.

Allergies

Allergies are exaggerated (hypersensitive) responses to certain antigens called *allergens*. The most common allergies involve antibodies of the IgE class. Hay fever, for instance, occurs when plasma cells secrete IgE antibodies specific for antigens on the surface of pollen grains **(Figure 43.26)**. Some IgE antibodies attach by their base to mast cells in connective tissues. Pollen grains that enter the body later attach to the antigen-binding sites of these IgE antibodies. Such attachment cross-links adjacent IgE molecules, inducing the mast cell to release histamine and other inflammatory chemicals. Acting on a variety of cell types, these chemicals bring about the typical allergy symptoms: sneezing, runny nose, teary eyes, and smooth muscle contractions in the lungs that can inhibit effective breathing. Drugs known as antihistamines block receptors for histamine, diminishing allergy symptoms (and inflammation).

Figure 43.26

Mast cells, IgE, and the allergic response.



In this example, pollen grains act as the allergen.

The process is as follows.

- IgE antibodies (Y-shaped) produced in response to initial exposure to an allergen bind to receptors on mast cells.
- On subsequent exposure to the same allergen, IgE molecules attached to a mast cell recognize and bind the allergen.
- Cross-linking of adjacent IgE molecules triggers release of histamine and other chemicals from vesicles inside the mast cell, leading to allergy symptoms.

An acute allergic response sometimes leads to a life-threatening reaction called *anaphylactic shock*. Inflammatory chemicals released from immune cells trigger constriction of bronchioles and sudden dilation of peripheral blood vessels, which causes a precipitous drop in blood pressure. Death may occur within minutes due to the inability to breathe and lack of blood flow. Substances that can cause anaphylactic shock in allergic individuals include bee venom, penicillin, peanuts, and shellfish. People with severe hypersensitivities often carry an autoinjector containing the hormone epinephrine. An injection of epinephrine rapidly counteracts this allergic response, constricting peripheral blood vessels, reducing swelling in the throat, and relaxing muscles in the lungs to help breathing (see Figure 45.20b).

Autoimmune Diseases

In some people, the immune system is active against particular molecules of the body, causing an **autoimmune disease**. Such a loss of self-tolerance has many forms. In systemic lupus erythematosus, commonly called *lupus*, the immune system generates antibodies against histones and DNA released by the normal breakdown of body cells. These self-reactive antibodies cause skin rashes, fever, arthritis, and kidney dysfunction. Other targets of autoimmunity include the insulin-producing beta cells of the pancreas (in type 1 diabetes) and the myelin sheaths that encase many neurons (in multiple sclerosis).

Heredity, sex, and environment all influence susceptibility to autoimmune disorders. For example, members of certain families show an increased susceptibility to particular autoimmune disorders. In addition, many autoimmune diseases affect females more often than males. Females are nine times as likely as males to have lupus and two to three times as likely to develop *rheumatoid arthritis*, a damaging and painful inflammation of the cartilage and bone in joints **(Figure 43.27)**. The causes of this sex bias, as well as the rise in autoimmune disease frequency in industrialized countries, are areas of active research and debate.

Figure 43.27

X-ray of hands that are deformed by rheumatoid arthritis.

An additional focus of current research on autoimmune disorders is the activity of *regulatory T cells*, nicknamed Tregs. These specialized T cells help modulate immune system activity and prevent response to self-antigens.

Exertion, Stress, and the Immune System

Many forms of exertion and stress influence immune system function. For example, moderate exercise improves immune system function and significantly reduces susceptibility to the common cold and other infections of the upper respiratory tract. In contrast, exercise to the point of exhaustion leads to more frequent infections and more severe symptoms. Psychological stress likewise disrupts immune system regulation, altering the interplay of the hormonal, nervous, and immune systems (see Figure 45.20). Research has also shown that rest is important for immunity: Adults who averaged fewer than 7 hours of sleep got sick three times as often when exposed to a cold virus as those who averaged at least 8 hours.

Immunodeficiency Diseases

A disorder in which an immune system response to antigens is defective or absent is called an immunodeficiency. Whatever its cause and nature, an immunodeficiency can lead to frequent and recurrent infections and increased susceptibility to certain cancers.

An *inborn immunodeficiency* results from a genetic or developmental defect in the production of immune system cells or of specific proteins, such as antibodies or the proteins of the complement system. Depending on the specific defect, either innate or adaptive defenses—or both—may be impaired. In severe combined immunodeficiency (SCID), functional lymphocytes are rare or absent. Lacking an adaptive immune response, people with SCID are susceptible to infections that can cause death in infancy, such as pneumonia and meningitis. Treatments include bone marrow and stem cell transplantation.

Later in life, exposure to chemicals or biological agents can cause an *acquired immunodeficiency*. Drugs used to fight autoimmune diseases or prevent transplant rejection suppress the immune system, leading to an immunodeficient state. Certain cancers also suppress the immune system, especially Hodgkin's disease, which damages the lymphatic system. Acquired immunodeficiencies range from temporary states that may arise from physiological stress to the devastating disease AIDS (acquired immune deficiency syndrome), which we'll explore in the next section.

Evolutionary Adaptations of Pathogens That Underlie Immune System Avoidance

EVOLUTION Just as immune systems that ward off pathogens have evolved in animals, mechanisms that thwart immune responses have evolved in pathogens. Using human pathogens as examples, we'll examine some common mechanisms: antigenic variation, latency, and direct attack on the immune system.

Antigenic Variation

One mechanism for escaping the body's defenses involves a pathogen altering how it appears to the immune system. If a pathogen changes the epitopes that it expresses to ones that a host has not previously encountered, it can reinfect or remain in the host without triggering the rapid and robust response mediated by memory cells. Such changes in epitope expression are called *antigenic variation*. The parasite that causes sleeping sickness (trypanosomiasis) provides an extreme example, periodically switching at random among 1,000 versions of the protein found over its entire surface. In the **Scientific Skills Exercise**, you will interpret data on this form of antigenic variation and the body's response.

Scientific Skills Exercise

Comparing Two Variables on a Common x-Axis**

AP Science Practices 3, 4, 6

How Does the Immune System Respond to a Changing Pathogen? Natural selection favors parasites that are able to maintain a low-level infection in a host for a long time. *Trypanosoma*, the unicellular parasite that causes sleeping sickness, is one example. The glycoproteins covering a trypanosome's surface are encoded by a gene that is duplicated more than 1,000 times in the organism's genome. Each copy is slightly different. By periodically switching among these genes, the trypanosome can display a series of surface glycoproteins with different molecular structures. In this exercise, you will interpret two data sets to explore hypotheses about the benefits to the trypanosome of its ever-shifting surface glycoproteins in avoiding the host's immune response.

Data from the Studies Study A measured the abundance of parasites in the blood of one human patient during the first few weeks of a chronic infection. The results are shown in the second column of the data table. Many decades after scientists first observed the pattern of *Trypanosoma* abundance over the course of

infection, researchers identified antibodies specific to different forms of the parasite's surface glycoprotein. The third and fourth columns of the table list the relative abundance of two such antibodies during the early period of chronic infection, using an index ranging from 0 (absent) to 1.

Day of infection	Study A: Millions of parasites/mL blood	Study B: Antibody specific to Variant A	Study B: Antibody specific to Variant B
4	0.1	0	0
6	0.3	0	0
8	1.2	0.2	0
10	0.2	0.5	0
12	0.2	1	0
14	0.9	1	0.1
16	0.6	1	0.3
18	0.1	1	0.9
20	0.7	1	1
22	1.2	1	1
24	0.2	1	1

Data from*L. J. Morrison et al., Probabilistic order in antigenic variation of Trypanosoma brucei, International Journal for Parasitology 35:961–972 (2005); and L. J. Morrison et al., Antigenic variation in the African trypanosome: molecular mechanisms and phenotypic complexity, Cellular Microbiology 1:1724–1734 (2009).*

Day of infection	Study A: Millions of parasites per milliliter of blood	Study B: Antibody specific to variant A	Study B: Antibody specific to variant B
4	0.1	0	0
6	0.3	0	0
8	1.2	0.2	0
10	0.2	0.5	0
12	0.2	1	0
14	0.9	1	0.1
16	0.6	1	0.3
18	0.1	1	0.9
20	0.7	1	1
22	1.2	1	1
24	0.2	1	1

Part A: INTERPRET THE DATA

- Which of the first two columns represents the independent variable and which the dependent variable? Plot the data from Study A as a line graph, putting the independent variable on the x-axis. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Visually displaying data in a graph can help make patterns in the data more noticeable. Describe any patterns revealed by your graph.
- Assume that a drop in parasite abundance reflects an effective immune response by the host. Formulate a hypothesis to explain the pattern you described in question 2.

Part B: INTERPRET THE DATA

- Note that the data for Study B were collected over the same period of infection (days 4–24) as the parasite abundance data in Study A. Therefore, you can incorporate these new data into your first graph, using the same x-axis. However, since the antibody level data are measured in a different way than the parasite abundance data, add a second set of y-axis labels on the right side of your graph. Then, using different colors or sets of symbols, add the data for the two antibody types. Labeling the y-axis two different ways enables you to compare how two dependent variables change relative to a shared independent variable.

- Describe any patterns you observe by comparing the two data sets over the same period. Do these patterns support your hypothesis from question 3? Do they prove it? Explain.
- Scientists can now also distinguish the abundance of trypanosomes recognized specifically by antibodies type A and type B. How would incorporating such information change your graph?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Antigenic variation is the main reason the influenza, or "flu," virus remains a major public health problem. As it replicates in one human host after another, the human flu virus undergoes frequent mutations. As a result, a new flu vaccine must be developed, produced, and distributed each year. In addition, the human flu virus occasionally forms new strains by exchanging genes with influenza viruses that infect domesticated animals, such as pigs or chickens. When this exchange of genes occurs, the new strain may not be recognized by any of the memory cells in the human population. The resulting outbreak can be deadly: The 1918–1919 influenza outbreak killed more than 20 million people. Similarly, the emergence of new variants of SARS-CoV-2 is of continuing concern.

Latency

Some viruses avoid an immune response by infecting cells and then entering a largely inactive state called *latency*. In latency, the production of most viral proteins and free viruses ceases; as a result, latent viruses do not trigger an adaptive immune response. Nevertheless, the viral genome persists in the nuclei of infected cells, either as a separate DNA molecule or as a copy integrated into the host genome. Latency typically persists until conditions arise that are favorable for viral transmission or unfavorable for host survival, such as when the host is infected by another pathogen. Such circumstances trigger the synthesis and release of free viruses that can infect new hosts.

Herpes simplex viruses provide a good example of latency. A number of these herpesviruses infect only humans and cause symptoms ranging from mild to life-threatening **(Table 43.1)**. The type 1 herpesvirus causes most oral herpes infections (often inaccurately called "cold" sores), whereas the sexually transmitted type 2 herpesvirus is responsible for most cases of genital herpes. People infected with either the type 1 or type 2 herpesvirus often have no symptoms. Instead, these viruses remain latent in certain neurons until a stimulus such as fever, emotional stress, or a hormonal change associated with the menstrual cycle reactivates the virus. Activation of the type 1 herpesvirus can result in the appearance of blisters around the mouth. Infections of the type 2 herpesvirus pose a serious threat to babies in utero during pregnancy and can increase transmission of HIV.

Table 43.1

Latency as a Shared Characteristic of Human Herpesviruses

Human Herpesvirus	Main Sites of Latency	Associated Diseases or Disorders
Herpes simplex virus-1 (HSV-1)	Clusters of neurons in spinal nerves	Cold sores
Herpes simplex virus-2 (HSV-2)	Clusters of neurons in spinal nerves	Genital ulcers
Varicella-zoster virus (VZV)	Clusters of neurons in spinal nerves	Chicken pox, shingles
Epstein-Barr virus (EBV)	Memory B cells	Some lymphoma, mononucleosis
Cytomegalovirus (CMV)	Monocytes and lymphocytes	Abnormal fetal development
Human herpesvirus 8 (HHV-8)	B cells	Kaposi's sarcoma

The table is as follows:

Table 43.1 Latency as a Shared Characteristic of Human Herpes Virus
Human herpes virus
Herpes simplex virus-1 (H S V- 1)
Herpes simplex virus-2 (H S V- 2)
Varicella-zoster virus (V Z V)
Epstein-Barr virus (E B V)
Cytomegalovirus (C M V)
Human herpesvirus 8 (H H V-8)

Attack on the Immune System: HIV

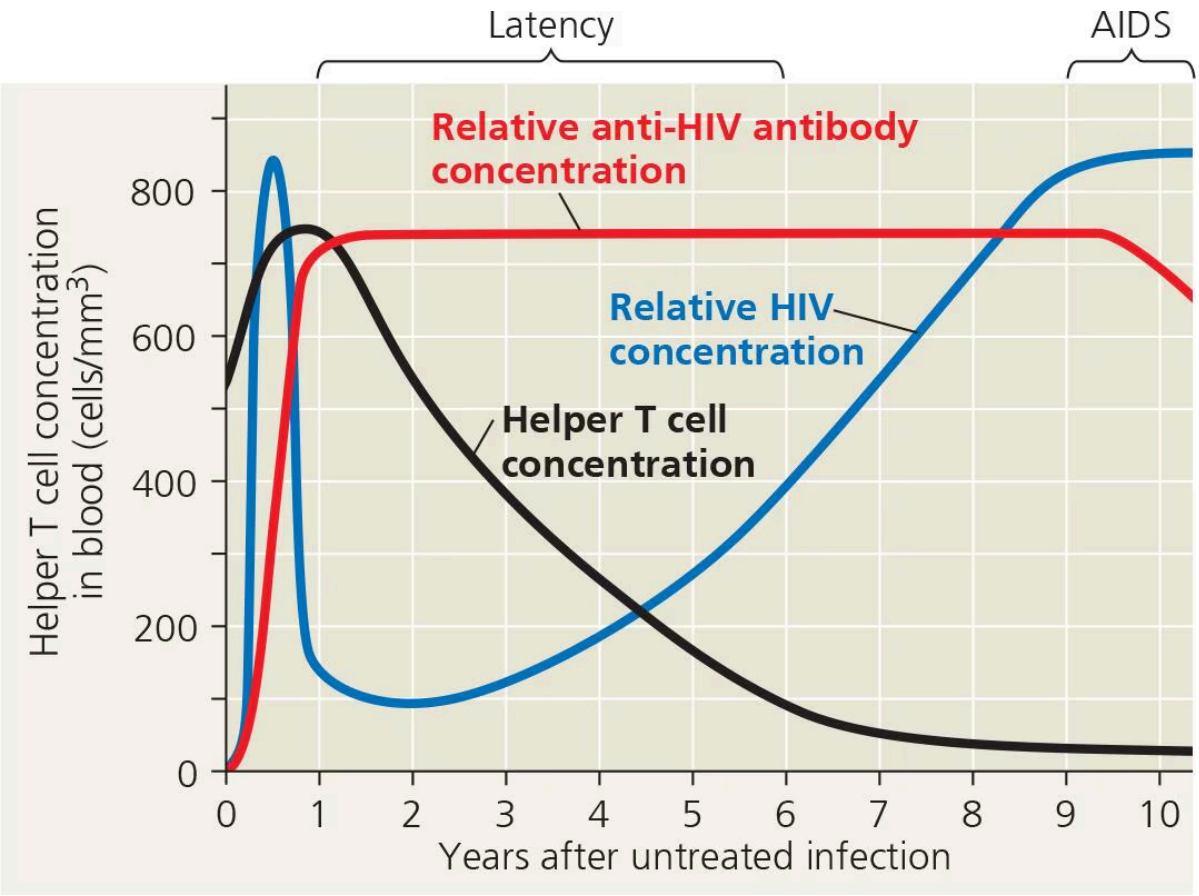
The **human immunodeficiency virus (HIV)**, the pathogen that causes AIDS, both escapes and attacks the adaptive immune response. HIV infects helper T cells with high efficiency by binding specifically to the CD4 accessory protein. HIV also infects some cell types that have low levels of CD4, such as macrophages and brain cells. Inside cells, the HIV RNA genome is reverse-transcribed, and the product DNA is integrated into the host cell's genome (see Figure 19.8). In this form, the viral genome can direct the production of new viruses.

Although the body responds to HIV with an immune response sufficient to eliminate most viral infections, some HIV invariably escapes. One reason HIV persists is that it has a very high mutation rate. Altered proteins on the surface of some mutated viruses reduce interaction with antibodies and cytotoxic T cells. Such viruses replicate and mutate further. HIV thus evolves within the body.

Over time, an untreated HIV infection not only avoids the adaptive immune response but also abolishes it **(Figure 43.28)**. Viral replication and cell death triggered by the virus lead to loss of helper T cells, impairing both humoral and cell-mediated immune responses. The eventual result is **acquired immunodeficiency syndrome(AIDS)**, an impairment in immune responses that leaves the body susceptible to infections and cancers that a healthy immune system would usually defeat. For example, *Pneumocystis jirovecii*, a common fungus that does not cause disease in healthy individuals, can result in severe pneumonia in people with AIDS. Such opportunistic diseases, as well as nerve damage and body wasting, are the primary causes of death from AIDS, rather than HIV itself.

Figure 43.28

The progress of an untreated HIV infection.



Graph of helper T cell concentration versus years after an untreated infection during the latency and AIDS periods, with lines for relative anti-HIV antibody concentration, relative HIV concentration, and helper T cell concentration. The x-axis is years after an untreated infection, ranging from 0 to 10 with increments every whole number. The y-axis is helper T cell concentration in blood (cells per millimeter cubed), ranging from 0 to 800 with increments every 100. The latency period is roughly 2 to 9 years post infection and AIDS develops 9 years post infection. Overall, helper T cell concentration is lowest as relative HIV concentration peaks (during AIDS), about 9 years after infection. Relative anti-HIV antibody concentration peaks during the latency 2 to 9 years after infection and then drops off as AIDS develops. Trend descriptions. Relative anti-HIV antibody concentration. Climbs steeply from 0 to almost 740 within the first year after an untreated infection, and then remains flat until about 9.5 years post infection, where it begins to fall. Helper T cell concentration. Climbs from about 550 to just over 700 in the first year post infection, and then drops to 100 by year 6 and about 50 by year 7, remaining at low levels through year 10. Relative HIV concentration. Climbs from 0 to about 850 in the first six months after infection, and then drops steeply to 100 during years 1-3, before climbing to about 850 by year 9, when AIDS develops.

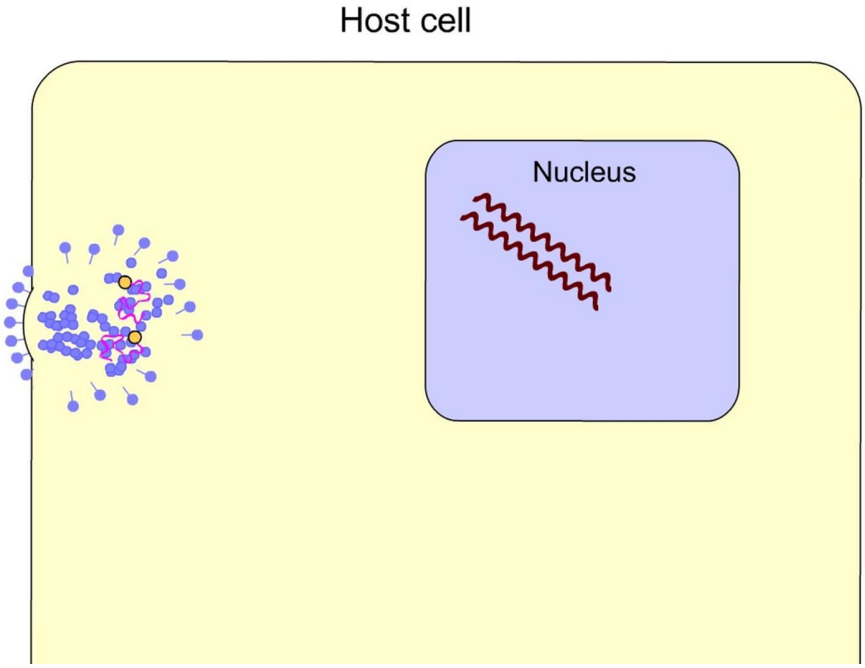
Interview

Interview with Flossie Wong-Staal: Characterizing HIV

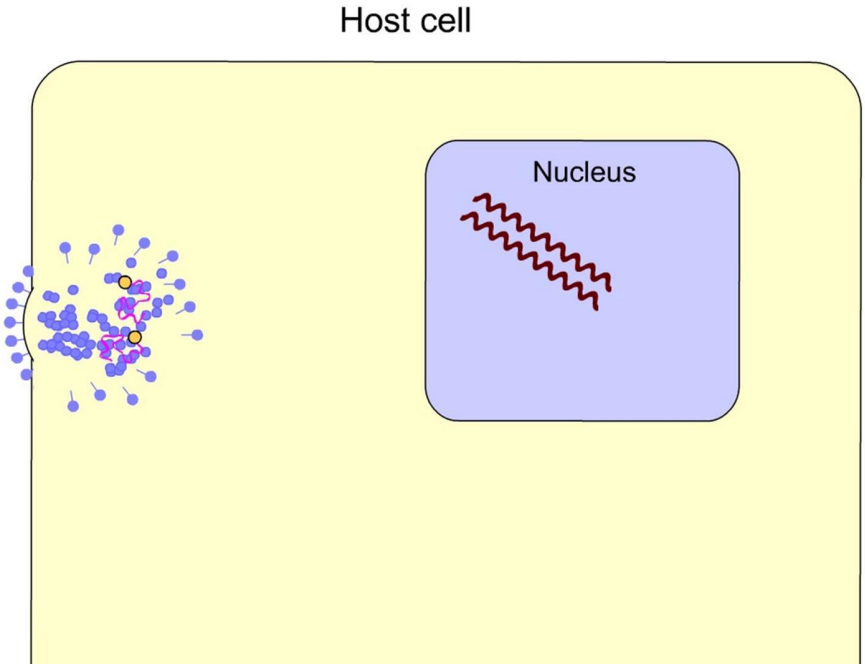
Transmission of HIV requires the transfer of virus particles or infected cells from person to person via body fluids such as semen, blood, or breast milk. Public health measures shown to reduce transmission include promoting the use of condoms during sex and supplying clean needles for intravenous drug users. Although no cure has been identified for HIV infection, drugs that slow HIV replication (known as anti-retroviral therapy, or ART) can reduce the risk of developing AIDS. By reducing the viral load, ART helps prevent transmission to others, including allowing HIV-positive individuals to give birth to babies without HIV.

Animation

HIV Reproductive Cycle



Play



Play

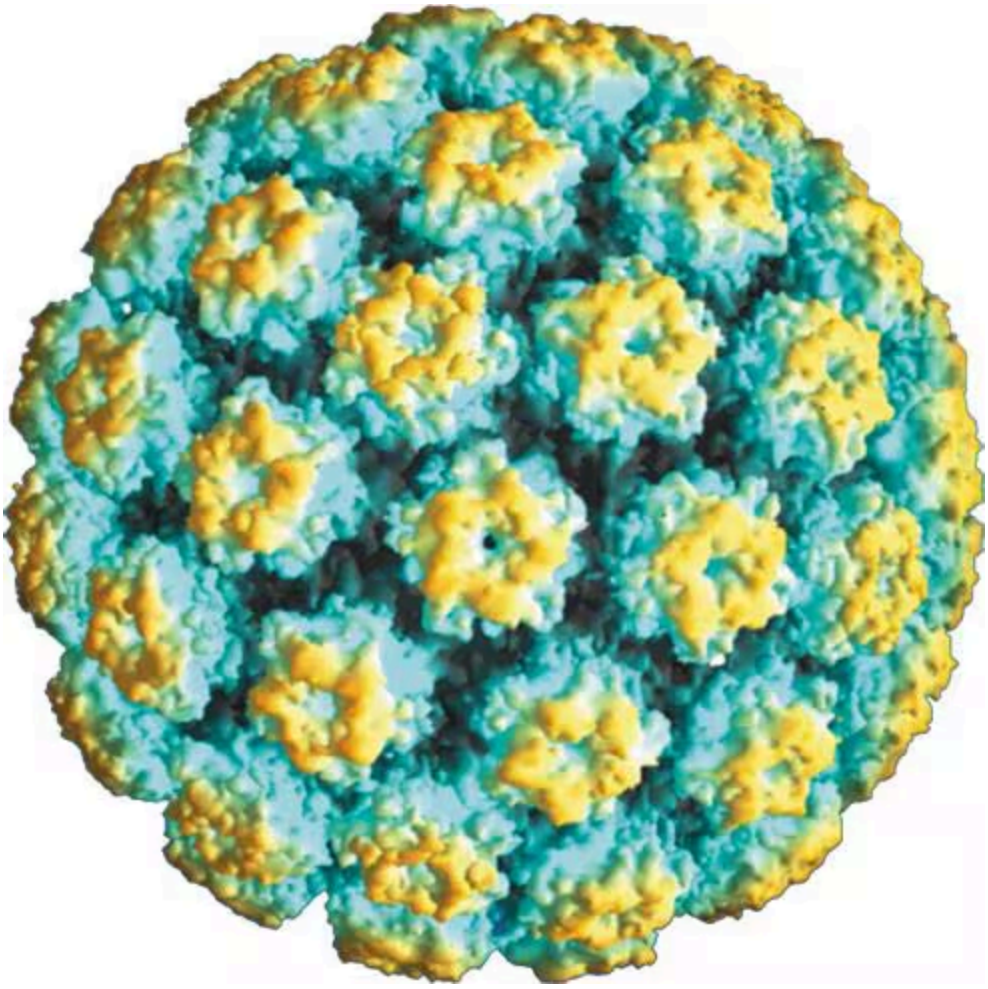
Cancer and Immunity

When adaptive immunity is inactivated, the frequency of certain cancers increases dramatically. For example, the risk of developing Kaposi's sarcoma is 20,000 times greater for patients with untreated AIDS than for people who are healthy. This observation was initially puzzling. If the immune system recognizes only nonself, it should fail to recognize the uncontrolled growth of self cells that is the hallmark of cancer. Recent studies suggest, however, that viruses are involved in about 10% of all human cancers. Because the immune system can recognize viral proteins as foreign, it can act as a defense against viruses that can cause cancer and against cancer cells that harbor viruses.

Scientists have identified numerous viruses that can cause cancer in humans. The Kaposi's sarcoma herpesvirus is one such virus (see Table 43.1). Hepatitis B virus, which can trigger liver cancer, is another. A vaccine introduced in 1986 for hepatitis B virus was the first vaccine shown to help prevent a specific human cancer. Rapid progress on developing vaccines for virus-induced cancers continues. In 2006, the release of a vaccine specific for human papillomavirus (HPV) marked a major victory against cervical cancer, as well as oral cancers that are becomingly increasingly common among men. The computer graphic image of an HPV particle in **Figure 43.29** illustrates the abundant copies of the capsid protein (yellow) that is used as the antigen in vaccination.

Figure 43.29

Human papillomavirus.



Interview

Interview with Harald zur Hausen: Discovering the cause of cervical cancer